

Data Cleaning and Preparation

```
In [ ]: import pandas as pd
```

```
In [ ]: df = pd.read_csv('SnapDeal.csv')
```

```
In [ ]: df.head()
```

	Timestamp	age	Gender	Purchase_Frequency	Purchase_Categories
0	2023/06/07 11:06:40 PM GMT+5:30	59	Others	Once a week	Beauty and Personal Care
1	2023/06/06 7:08:47 PM GMT+5:30	48	Female	Once a month	Beauty and Personal Care;Clothing and Fashion;...
2	2023/06/08 9:39:30 PM GMT+5:30	57	Prefer not to say	Multiple times a week	Groceries and Gourmet Food
3	2023/06/07 9:22:24 PM GMT+5:30	30	Others	Few times a month	Groceries and Gourmet Food;Beauty and Personal...
4	2023/06/06 7:21:26 PM GMT+5:30	51	Female	Once a month	Groceries and Gourmet Food;Beauty and Personal...

5 rows × 24 columns



```
In [ ]: df.shape
```

(800, 24)

In []: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Timestamp                                800 non-null    object
1   age                                       800 non-null    int64
2   Gender                                   800 non-null    object
3   Purchase_Frequency                     800 non-null    object
4   Purchase_Categories                    800 non-null    object
5   Personalized_Recommendation_Frequency 800 non-null    object
6   Browsing_Frequency                     800 non-null    object
7   Product_Search_Method                  651 non-null    object
8   Search_Result_Exploration              800 non-null    object
9   Customer_Reviews_Importance            800 non-null    int64
10  Add_to_Cart_Browsing                    800 non-null    object
11  Cart_Completion_Frequency              800 non-null    object
12  Cart_Abandonment_Factors                800 non-null    object
13  Saveforlater_Frequency                  800 non-null    object
14  Review_Left                             800 non-null    object
15  Review_Reliability                     800 non-null    object
16  Review_Helpfulness                     800 non-null    object
17  Personalized_Recommendation_Frequency 800 non-null    int64
18  Recommendation_Helpfulness              800 non-null    object
19  Rating_Accuracy                        800 non-null    int64
20  Shopping_Satisfaction                   800 non-null    int64
21  Service_Appreciation                    800 non-null    object
22  Improvement_Areas                       800 non-null    object
23  transaction                             800 non-null    int64
dtypes: int64(6), object(18)
memory usage: 150.1+ KB
```

No Duplicates Found

In []: df.duplicated().sum()

```
np.int64(0)
```

Standardizing categorical / numerical entries

In []: non_numeric_cols = df.select_dtypes(include=['object']).columns.to_list()

```
In [ ]: # Removing extra spaces from non numeric columns
        for col in non_numeric_cols:
            df[col] = df[col].str.strip()
```

```
In [ ]: df['Gender'].unique()

        array(['Others', 'Female', 'Prefer not to say', 'Male'],
              dtype=object)
```

```
In [ ]: df['Purchase_Frequency'].unique()

        array(['Once a week', 'Once a month', 'Multiple times a week',
              'Few times a month', 'Less than once a month'], dtype=object)
```

```
In [ ]: df['Personalized_Recommendation_Frequency '].unique()

        array([1, 5, 4, 3, 2])
```

```
In [ ]: df['Personalized_Recommendation_Frequency'].unique()

        array(['Yes', 'Sometimes', 'No'], dtype=object)
```

```
In [ ]: df['Recommendation_Helpfulness'].unique()

        array(['Yes', 'No', 'Sometimes'], dtype=object)
```

```
In [ ]: # Removing duplicate column
        df.drop(columns=['Personalized_Recommendation_Frequency'], inplace=True)
```

```
In [ ]: # Missing Values
df.isna().sum()
```

	0
Timestamp	0
age	0
Gender	0
Purchase_Frequency	0
Purchase_Categories	0
Browsing_Frequency	0
Product_Search_Method	149
Search_Result_Exploration	0
Customer_Reviews_Importance	0
Add_to_Cart_Browsing	0
Cart_Completion_Frequency	0
Cart_Abandonment_Factors	0
Saveforlater_Frequency	0
Review_Left	0
Review_Reliability	0
Review_Helpfulness	0
Personalized_Recommendation_Frequency	0
Recommendation_Helpfulness	0
Rating_Accuracy	0
Shopping_Satisfaction	0
Service_Appreciation	0
Improvement_Areas	0
transaction	0

dtype: int64

```
In [ ]: # Column Product_Search_Method has some null values
df['Product_Search_Method'].unique()
```

```
array(['others', 'categories', 'Keyword', 'Filter', nan],
      dtype=object)
```

```
In [ ]: df['Product_Search_Method'].value_counts()
```

	count
Product_Search_Method	
others	182
Keyword	163
categories	156
Filter	150

dtype: int64

```
In [ ]: # New category added as Unknown Search Method
df['Product_Search_Method'].fillna('Unknown', inplace=True)
```

/tmp/ipykernel_2675/3727337012.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Product_Search_Method'].fillna('Unknown', inplace=True)
```

```
In [ ]: df['Product_Search_Method'].unique()
```

```
array(['others', 'categories', 'Keyword', 'Filter', 'Unknown'],
      dtype=object)
```

```
In [ ]: # Fixing Col Names
df.columns = df.columns.str.strip()
```

```
In [ ]: df.columns
```

```
Index(['Timestamp', 'age', 'Gender', 'Purchase_Frequency',  
      'Purchase_Categories', 'Browsing_Frequency',  
      'Product_Search_Method',  
      'Search_Result_Exploration', 'Customer_Reviews_Importance',  
      'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',  
      'Cart_Abandonment_Factors', 'Saveforlater_Frequency',  
      'Review_Left',  
      'Review_Reliability', 'Review_Helpfulness',  
      'Personalized_Recommendation_Frequency',  
      'Recommendation_Helpfulness',  
      'Rating_Accuracy', 'Shopping_Satisfaction',  
      'Service_Appreciation',  
      'Improvement_Areas', 'transaction'],  
      dtype='object')
```

```
In [ ]: # All numeric cols have correct data type  
df['Customer_Reviews_Importance'].dtype
```

```
dtype('int64')
```

```
In [ ]: df['Shopping_Satisfaction'].dtype
```

```
dtype('int64')
```

```
In [ ]: df['Rating_Accuracy'].dtype
```

```
dtype('int64')
```

```
In [ ]: df['age'].dtype
```

```
dtype('int64')
```

```
In [ ]: df.to_csv('cleaned_data.csv')
```

```
In [ ]:
```